

Primitive Adjacent Sums in Finite Fields

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Abstract

We prove the Hou-Sun conjecture that, for every prime power $q = p^n > 7$, there exists a cyclic ordering of $\mathbb{F}_q$ whose adjacent sums are primitive. Moreover, we prove a stronger result. The adjacent sums take at most $n + 2$ distinct values in odd characteristic, and exactly n in characteristic two. In particular, when $p \geq 5$ and n is even, or when $p = 3$ and $3 \mid n$, one can attain $n + 1$ distinct values and the bound is optimal, while for prime fields of order greater than seven, the minimum is three. Given a representation of $\mathbb{F}_q$, we also give a deterministic algorithm that constructs such an ordering in time polynomial in q .

Keywords: finite fields; primitive elements; cyclic orderings; Hamiltonian cycles; affine spanning.

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1 Introduction

Let $q = p^n$, where p is prime and $n \geq 1$, and regard $\mathbb{F}_q$ as an n -dimensional vector space over $\mathbb{F}_p$. The multiplicative group $\mathbb{F}_q^\times$ is cyclic of order $q - 1$; its generators are called primitive elements. We write

$$\mathcal{P}_q = \{z \in \mathbb{F}_q^\times : \text{ord}(z) = q - 1\}.$$

A cyclic ordering of $\mathbb{F}_q$ is a sequence $A = (a_0, \dots, a_{q-1})$ containing each field element exactly once. We denote its set of adjacent sums by

$$T(A) = \{a_j + a_{j+1} : 0 \leq j < q\}, \quad a_q = a_0.$$

Sun [1, Conjecture 3.5(i)] stated the following conjecture, proposed jointly with Hou: for every $q > 7$, there is a cyclic ordering A of $\mathbb{F}_q$ such that $T(A) \subseteq \mathcal{P}_q$. Equivalently, the graph with vertex set $\mathbb{F}_q$, in which distinct vertices x, y are joined when $x + y \in \mathcal{P}_q$, has a Hamiltonian cycle.

She [2, Theorem 1.2(1) and Remark 1.3] proved the conjecture for $q > 18\,888\,871$ and, over prime fields, for $p > 61$.

Our proof in odd characteristic has three ingredients. First, we construct a nondegenerate three-term arithmetic progression $c - d, c, c + d$ of primitive elements. Second, a character-sum argument shows that the primitive elements affinely span $\mathbb{F}_q$ over $\mathbb{F}_p$. Third, we construct Hamiltonian cycles in $\mathbb{F}_p^n$ whose edge sums belong to a small prescribed set. An affine change of coordinates then transfers these cycles to $\mathbb{F}_q$.

In characteristic two, the argument is simpler: a primitive normal basis together with a cyclic binary Gray code gives the required ordering.

1.1 Main results

Theorem 1.1 (Primitive adjacent sums). For every prime power $q = p^n > 7$, there exists a cyclic ordering $A = (a_0, \dots, a_{q-1})$ of $\mathbb{F}_q$ such that

$$a_j + a_{j+1} \in \mathcal{P}_q \quad (0 \leq j < q), \quad a_q = a_0, \quad (1.1)$$

and

$$|T(A)| \leq \begin{cases} n + 2, & p \text{ is an odd prime,} \\ n, & p = 2. \end{cases} \quad (1.2)$$

When $p \geq 5$ and n is even, or when $p = 3$ and $3 \mid n$, one can achieve $|T(A)| = n + 1$, the minimum possible value. In characteristic two, one can achieve $|T(A)| = n$, also the minimum possible value. For prime fields of order greater than seven, the minimum is $|T(A)| = 3$, and it is attained by the construction.

The lower bounds follow from the spanning properties of the adjacent-sum set. For any cyclic ordering A in odd characteristic, $T(A)$ affinely spans $\mathbb{F}_p^n$ and contains at least three elements. Thus

$$|T(A)| \geq \max\{n + 1, 3\}. \quad (1.3)$$

In characteristic two, the adjacent-sum set of any cyclic ordering linearly spans the whole space, so $|T(A)| \geq n$. We prove these assertions in Section 6.

The construction also applies to prescribed adjacent-sum sets in vector spaces. For a nonempty set $S \subseteq V = \mathbb{F}_p^n$, define its affine hull by

$$\text{Aff}(S) = c + \text{span}_{\mathbb{F}_p}(S - c), \quad c \in S.$$

This definition is independent of the choice of c ; we set $\text{Aff}(\emptyset) = \emptyset$. We say that S affinely spans V if $\text{Aff}(S) = V$, and linearly spans V if $\text{span}_{\mathbb{F}_p}(S) = V$. In odd characteristic, a nondegenerate three-term arithmetic progression is a triple $c - d, c, c + d$ with $d \neq 0$.

Theorem 1.2 (Prescribed adjacent-sum sets). Let $V = \mathbb{F}_p^n$, where $n \geq 1$ and $|V| > 2$, and let $S \subseteq V$.

Suppose that p is odd. If S affinely spans V and contains a nondegenerate three-term arithmetic progression, then V has a cyclic ordering whose adjacent sums belong to S and take at most $n+2$ distinct values. Conversely, the existence of a cyclic ordering of V with all adjacent sums in S implies that S affinely spans V .

When $p \geq 5$ and n is even, or when $p = 3$ and $3 \mid n$, V has a cyclic ordering with all adjacent sums in S if and only if S affinely spans V . In this case, exactly $n+1$ distinct adjacent sums suffice, and this number is optimal.

When $p = 2$, V has a cyclic ordering with all adjacent sums in S if and only if S linearly spans V . In this case, exactly n distinct adjacent sums suffice, and this number is optimal.

Given an irreducible-polynomial representation of $\mathbb{F}_q$, our constructions yield a deterministic algorithm with running time polynomial in q . In odd characteristic, once the complete prime factorization of $q-1$ is given, the algorithm uses $O(q(\log q)^2 + qn^2)$ arithmetic operations, counting operations in $\mathbb{F}_q$ and $\mathbb{F}_p$ at unit cost. Section 7 describes the algorithm and its complexity.

2 Three-term arithmetic progressions of primitive elements

Let q be odd and put

$$N = q - 1, \quad \vartheta = \frac{\varphi(N)}{N}, \quad W = 2^{\omega(N)}, \quad (2.1)$$

where φ is Euler's totient function and $\omega(N)$ is the number of distinct prime divisors of N . For a generator g of $\mathbb{F}_q^\times$,

$$\mathcal{P}_q = \{g^k : \gcd(k, N) = 1\}, \quad |\mathcal{P}_q| = \varphi(N) = \vartheta N.$$

The number W counts the positive squarefree divisors of N .

Throughout the paper, squares and cubes mean elements of $(\mathbb{F}_q^\times)^2$ and $(\mathbb{F}_q^\times)^3$, respectively. If $3 \mid N$, then g^k is a cube if and only if $3 \mid k$; if $3 \nmid N$, every nonzero element is a cube.

We first construct nonzero squares r, s satisfying $r + s = 2$. Multiplying $r, 1, s$ by a suitable common factor then gives a three-term arithmetic progression of primitive elements.

Lemma 2.1. Let q be odd, and let nonzero squares $r, s \in \mathbb{F}_q$ satisfy

$$r + s = 2, \quad r \neq s. \quad (2.2)$$

If $3 \mid q - 1$, assume also that at least one of $r, s, r/s$ is a cube. Then there exists $c \in \mathbb{F}_q^\times$ such that cr, c, cs are all primitive.

Proof. Fix a generator g of the multiplicative group and write

$$r = g^k, \quad s = g^\ell, \quad c = g^t.$$

The three elements are primitive if and only if

$$\gcd(t + k, N) = \gcd(t, N) = \gcd(t + \ell, N) = 1, \quad (2.3)$$

or, equivalently, for each prime divisor $v \mid N$,

$$t \not\equiv 0, -k, -\ell \pmod{v}. \quad (2.4)$$

For $v = 2$, the square condition gives $k, \ell \equiv 0 \pmod{2}$, so we may take $t \equiv 1 \pmod{2}$. For $v = 3$, the cube condition gives $k \equiv 0, \ell \equiv 0$, or $k \equiv \ell \pmod{3}$, so at most two residue classes are

forbidden. For $v \geq 5$, at most three residue classes are forbidden, leaving at least one admissible class. Choose an admissible class a_v for each v . By the Chinese remainder theorem, we can solve

$$t \equiv a_v \pmod{v} \quad (v \mid N).$$

Choose an integer solution and set $c = g^t$. This choice satisfies (2.3). $\square$

Define

$$d = \frac{c(s-r)}{2}. \quad (2.5)$$

Since $c \neq 0$, $r \neq s$, and the characteristic is odd, we have $d \neq 0$, and

$$c-d = c\left(1 - \frac{s-r}{2}\right) = cr, \quad c+d = c\left(1 + \frac{s-r}{2}\right) = cs. \quad (2.6)$$

Thus $c-d, c, c+d$ form a nondegenerate three-term arithmetic progression of primitive elements.

We first consider the case $3 \nmid q-1$.

A multiplicative character is a group homomorphism $\xi : \mathbb{F}_q^\times \rightarrow \mathbb{C}^\times$. We denote the principal character by ξ_0 and extend every multiplicative character to zero by

$$\xi(0) = 0, \quad \xi_0(z) = \begin{cases} 1, & z \neq 0, \\ 0, & z = 0. \end{cases}$$

For odd q , let χ be the quadratic character. For $z \neq 0$, we have $\chi(z) = 1$ if and only if z is a square. For a nonprincipal multiplicative character ξ , character orthogonality gives

$$\sum_{z \in \mathbb{F}_q^\times} \xi(z) = 0. \quad (2.7)$$

Lemma 2.2. Let $q > 9$ be odd, and suppose that $3 \nmid q-1$. Then there exist nonzero squares r, s satisfying (2.2).

Proof. The number of ordered pairs of nonzero squares satisfying $r+s=2$ is

$$M = \frac{1}{4} \sum_{r \neq 0,2} (1 + \chi(r))(1 + \chi(2-r)). \quad (2.8)$$

By character orthogonality,

$$\sum_{r \neq 0,2} \chi(r) = \sum_{r \neq 0,2} \chi(2-r) = -\chi(2).$$

The equation $v^2 = u^2 - 1$ is equivalent to $(u-v)(u+v) = 1$. Its solutions are parametrized by $w \in \mathbb{F}_q^\times$:

$$u = \frac{w + w^{-1}}{2}, \quad v = \frac{w^{-1} - w}{2}.$$

There are therefore $q-1$ solutions, so

$$q + \sum_u \chi(u^2 - 1) = q - 1.$$

The substitution $r = 1 + u$ gives

$$\sum_r \chi(r(2-r)) = -\chi(-1).$$

Expanding (2.8) gives

$$M = \frac{q-2-2\chi(2)-\chi(-1)}{4} \geq \frac{q-5}{4} > 1. \quad (2.9)$$

The only pair with $r = s$ is $(1, 1)$. Removing it leaves a pair with distinct entries. $\square$

It remains to treat the case $3 \mid q - 1$. Then $6 \mid q - 1$ and $p \neq 3$. Choose a generator g of the multiplicative group and an element ζ of order three. Then

$$\zeta^2 + \zeta + 1 = 0, \quad (2\zeta + 1)^2 = -3,$$

and hence

$$\chi(3) = \chi(-1). \quad (2.10)$$

Also, $-1 = g^{N/2}$ is a cube, and ζ, ζ^2 are squares. We use the choices in Table 1, where integers denote elements of the prime field.

Table 1: Choices of r, s when $3 \mid q - 1$

Hypotheses	r	s	Cube condition
$\chi(-1) = 1$	-1	3	r is a cube
$\chi(-1) = \chi(2) = -1$, and 2 is a cube	4	-2	r is a cube
$\chi(-1) = \chi(2) = -1$, and 2 is not a cube	-2ζ	$-2\zeta^2$	At least one of $r, s, r/s$ is a cube
$\chi(-1) = -1$, $\chi(2) = 1$, and $p \neq 7$	8	-6	r is a cube

In the first row, (2.10) gives $\chi(3) = 1$, and $-1 \neq 3$. In the second row, $\chi(-2) = \chi(-1)\chi(2) = 1$, and 4 is a cube; the equality $4 = -2$ would imply $p \mid 6$. In the fourth row, $8 = 2^3$ and

$$\chi(8) = 1, \quad \chi(-6) = \chi(-1)\chi(2)\chi(3) = 1;$$

the equality $8 = -6$ would imply $p \mid 14$. The parameters in these three rows are nonzero and distinct, and their sum is two.

In the third row, $r + s = -2(\zeta + \zeta^2) = 2$, both parameters are nonzero squares, and $r \neq s$ follows from $\zeta \neq \zeta^2$. Let η be a multiplicative character of order three, whose kernel is the subgroup of cubes. If $\eta(\zeta) = 1$, then $\eta(r/s) = 1$. If $a = \eta(\zeta) \neq 1$, the assumption that two is not a cube gives $b = \eta(2) \in \{a, a^2\}$, and

$$\eta(r) = ba, \quad \eta(s) = ba^2.$$

When $b = a$, we have $\eta(s) = 1$; when $b = a^2$, we have $\eta(r) = 1$. Thus the cube condition holds in either case.

Lemma 2.3 (Weil bound [5, Theorem 5.41]). Let q be odd, let $f \in \mathbb{F}_q[X]$ be a squarefree polynomial of degree $D > 0$, and let χ be the quadratic character with $\chi(0) = 0$. Then

$$\left| \sum_{x \in \mathbb{F}_q} \chi(f(x)) \right| \leq (D - 1)\sqrt{q}. \quad (2.11)$$

Here squarefree means $\gcd(f, f') = 1$. The general bound for multiplicative character sums is also stated in the preprint version of [4, Lemma 2.2].

It remains to consider characteristic seven with $\chi(-1) = -1$. Since $3^2 = 2$, we have $\chi(2) = 1$; since $7 \equiv -1 \pmod{4}$, these fields have orders $q = 7^n$ with n odd. The condition $q > 7$ gives $n \geq 3$ and $q \geq 343$. Consider the curve

$$y^2 = 2 - x^6. \quad (2.12)$$

In characteristic seven, $f(X) = 2 - X^6$ satisfies $f'(X) = -6X^5$ and $f(0) = 2 \neq 0$, so f is squarefree. Lemma 2.3 gives

$$\left| \sum_x \chi(2 - x^6) \right| \leq 5\sqrt{q}.$$

Hence the number of affine points on the curve satisfies

$$T = q + \sum_x \chi(2 - x^6) \geq q - 5\sqrt{q} > 20. \quad (2.13)$$

At most 2, 12, 6 points satisfy $x = 0$, $x^6 = 1$, or $y = 0$, respectively. Removing these points leaves a point with $x \neq 0$, $x^6 \neq 1$, and $y \neq 0$. Set

$$r = x^6 = (x^3)^2 = (x^2)^3, \quad s = y^2.$$

Then r, s are nonzero squares, r is a cube, and $r + s = 2$. If $r = s$, then $r = 1$, contradicting $x^6 \neq 1$.

Proposition 2.4 (Three-term arithmetic progressions of primitive elements). For every odd $q > 7$ with $q \neq 9$, there exist $c, d \in \mathbb{F}_q$ such that

$$d \neq 0, \quad c - d, c, c + d \in \mathcal{P}_q. \quad (2.14)$$

Proof. If $3 \nmid q-1$, Lemma 2.2 supplies the required parameters. If $3 \mid q-1$, use the choices above. Lemma 2.1 then gives a common factor, and choosing d as in (2.5)–(2.6) gives the required triple. $\square$

3 The affine span of the primitive elements

We prove that the primitive elements affinely span $\mathbb{F}_q$ over $\mathbb{F}_p$, using their character indicator and a Gauss-sum estimate. We first bound the number of squarefree divisors of $q-1$.

Lemma 3.1. If $q = p^n > 7$, where p is an odd prime and $n \geq 2$, then

$$4^{\omega(q-1)} < q. \quad (3.1)$$

Proof. Let $r = \omega(q-1) \geq 1$, and denote the j th prime by π_j . When $r \geq 5$,

$$q-1 \geq \prod_{j=1}^r \pi_j > 4^r,$$

where the second inequality follows from $2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 = 2310 > 4^5$ and $\pi_j > 4$ for $j > 5$. When $r = 1$, we have $q \geq 9 > 4$.

For $r = 2, 3, 4$, suppose that $q \leq 4^r$. The lower bound $q-1 \geq \prod_{j=1}^r \pi_j$ restricts the possible field orders to the ranges in Table 2.

Table 2: Candidate field orders for $r = 2, 3, 4$

Assumed r	Range of q	Odd prime powers p^n in this range, $n \geq 2$	Value of $\omega(q-1)$
2	$7 \leq q \leq 16$	9	$\omega(8) = 1$
3	$31 \leq q \leq 64$	49	$\omega(48) = 2$
4	$211 \leq q \leq 256$	243	$\omega(242) = 2$

To enumerate these prime powers, it suffices to check the primes $p \leq \sqrt{q}$. The candidates satisfy

$$9 - 1 = 2^3, \quad 49 - 1 = 2^4 \cdot 3, \quad 243 - 1 = 2 \cdot 11^2,$$

respectively. Each contradicts the assumed value of r . $\square$

Let μ denote the Möbius function. The multiplicative character group is cyclic of order N , so it has exactly $\varphi(d)$ characters of order d for each $d \mid N$.

Lemma 3.2 (Character expansion of the primitive-element indicator [4, Equation (1) of the preprint]). For every $z \in \mathbb{F}_q$,

$$\mathbf{1}_{\mathcal{P}_q}(z) = \vartheta \sum_{d|N} \frac{\mu(d)}{\varphi(d)} \sum_{\text{ord}(\xi)=d} \xi(z). \quad (3.2)$$

The inner sum runs over the multiplicative characters of order d . Taking $n = 1$ and $e = N$ in the cited formula gives this identity. By our convention for characters at zero, both sides vanish at $z = 0$.

Primitive elements in affine hyperplanes were studied by Cohen [6].

An additive character is a group homomorphism $\psi : (\mathbb{F}_q, +) \rightarrow \mathbb{C}^\times$. For a nontrivial additive character, the orthogonality relation is

$$\sum_{z \in \mathbb{F}_q} \psi(z) = 0. \quad (3.3)$$

Lemma 3.3 (Gauss sums [5, Theorem 5.11]). Let ξ be a nonprincipal multiplicative character and ψ a nontrivial additive character, and set

$$G(\xi, \psi) = \sum_{x \in \mathbb{F}_q^\times} \xi(x) \psi(x).$$

Then

$$|G(\xi, \psi)| = \sqrt{q}. \quad (3.4)$$

For the principal multiplicative character, (3.3) gives

$$G(\xi_0, \psi) = \sum_{x \neq 0} \psi(x) = -1. \quad (3.5)$$

Proposition 3.4 (Affine span of the primitive elements). For every odd $q = p^n > 7$,

$$\text{Aff}(\mathcal{P}_q) = \mathbb{F}_q. \quad (3.6)$$

Proof. When $n = 1$, the elements g, g^{-1} are distinct and primitive, so their affine hull is the whole one-dimensional space. We may therefore assume that $n \geq 2$. Lemma 3.1 gives

$$W < \sqrt{q}. \quad (3.7)$$

For any nontrivial additive character ψ , Lemma 3.3 and (3.5) give

$$|G(\xi, \psi)| = \sqrt{q} \quad (\xi \neq \xi_0), \quad G(\xi_0, \psi) = -1. \quad (3.8)$$

Multiplying (3.2) by $\psi(z)$ and summing yields

$$\sum_{z \in \mathcal{P}_q} \psi(z) = -\vartheta + \vartheta \sum_{\substack{d|N \\ d>1}} \frac{\mu(d)}{\varphi(d)} \sum_{\text{ord}(\xi)=d} G(\xi, \psi).$$

For each squarefree divisor $d > 1$, the inner sum contains $\varphi(d)$ characters. Their total contribution has absolute value at most $\vartheta \sqrt{q}$. Thus

$$\left| \sum_{z \in \mathcal{P}_q} \psi(z) \right| \leq \vartheta(1 + (W - 1)\sqrt{q}) < \vartheta(q - 1) = |\mathcal{P}_q|, \quad (3.9)$$

where the strict inequality follows from

$$1 + (W - 1)\sqrt{q} < q + 1 - \sqrt{q} < q - 1.$$

If $\mathcal{P}_q$ were contained in a proper affine subspace $a_0 + U$, choose a nonzero linear functional $L : \mathbb{F}_q \rightarrow \mathbb{F}_p$ such that $L(U) = 0$. Let $a = L(a_0)$. Then

$$\psi_L(z) = \exp\left(\frac{2\pi i L(z)}{p}\right)$$

is a nontrivial additive character and has the constant value $\exp(2\pi i a/p)$ on $\mathcal{P}_q$. Hence

$$\left| \sum_{z \in \mathcal{P}_q} \psi_L(z) \right| = |\mathcal{P}_q|,$$

contradicting (3.9). □

Corollary 3.5. For every $c \in \mathcal{P}_q$,

$$\text{span}_{\mathbb{F}_p}(\mathcal{P}_q - c) = \mathbb{F}_q. \quad (3.10)$$

If $b_1 \in \mathcal{P}_q - c$ and $b_1 \neq 0$, one can choose $b_2, \dots, b_n$ from $\mathcal{P}_q - c$ so that $b_1, \dots, b_n$ form an $\mathbb{F}_p$ -basis.

Proof. The affine-hull identity

$$\text{Aff}(\mathcal{P}_q) = c + \text{span}_{\mathbb{F}_p}(\mathcal{P}_q - c)$$

and Proposition 3.4 give (3.10). If linearly independent vectors $b_1, \dots, b_j$ have been selected, with $j < n$, then (3.10) gives a vector

$$b_{j+1} \in (\mathcal{P}_q - c) \setminus \text{span}(b_1, \dots, b_j).$$

Repeating this step gives the required basis. □

For c, d from Proposition 2.4, set $b_1 = d$ and apply Corollary 3.5 to obtain

$$c \in \mathcal{P}_q, \quad c \pm b_1 \in \mathcal{P}_q, \quad c + b_j \in \mathcal{P}_q \quad (2 \leq j \leq n).$$

To construct this basis, inspect the primitive elements z in a fixed order, retaining each element for which $z - c$ increases the rank. Section 7 uses this procedure in the general construction.

4 Hamiltonian cycles with prescribed adjacent sums

Let p be an odd prime, let $n \geq 1$, and let $e_1, \dots, e_n$ denote the standard basis of $V = \mathbb{F}_p^n$. Put

$$\Lambda_n = \{0, e_1, -e_1, e_2, \dots, e_n\}. \quad (4.1)$$

For distinct vertices x, y , the *sum label* of the edge $\{x, y\}$ is $x + y$. A *zero-sum edge* has sum label zero. For a cyclic ordering, the sum labels are its adjacent sums. We construct cyclic orderings with all sum labels in Λ_n and with a zero-sum edge.

Theorem 4.1 (Cyclic orderings with prescribed sums). For every odd prime p and every integer $n \geq 1$, there exists a cyclic ordering

$$(x_0, \dots, x_{p^n-1})$$

of $V = \mathbb{F}_p^n$ such that every adjacent sum, including the closing sum, belongs to Λ_n , and at least one edge has sum zero.

4.1 A product construction

Lemma 4.2 (Product construction). Let U and W be finite vector spaces of odd order. Suppose that they have cyclic orderings whose adjacent sums belong to $S_U \subseteq U$ and $S_W \subseteq W$, respectively, and that each ordering contains an edge with sum zero. Then $U \oplus W$ has a cyclic ordering containing an edge with sum zero and having all adjacent sums in

$$(S_U \times \{0\}) \cup (\{0\} \times S_W). \quad (4.2)$$

Proof. Let $R = |U|$ and $T = |W|$. Delete an edge with sum zero from the cyclic ordering of U to obtain the spanning path

$$(h_0, \dots, h_{R-1}), \quad h_0 = u, \quad h_{R-1} = -u.$$

Fix a cyclic ordering $(w_0, \dots, w_{T-1})$ of W , with indices taken modulo T . For each w_j , define the segment with R vertices

$$Q_{w_j} = ((h_i, (-1)^i w_j))_{i=0}^{R-1}. \quad (4.3)$$

Concatenate these segments in the order $j = 0, \dots, T-1$ and join the last vertex to the first.

For each fixed i , the elements $(-1)^i w_j$ run through W as j varies, while distinct indices i give distinct first coordinates h_i . Thus every vertex of $U \oplus W$ occurs exactly once.

The adjacent sums within each segment satisfy

$$(h_i, (-1)^i w_j) + (h_{i+1}, (-1)^{i+1} w_j) = (h_i + h_{i+1}, 0), \quad 0 \leq i < R-1.$$

These sums belong to $S_U \times \{0\}$.

Since R is odd, $R-1$ is even. The first and last vertices of each segment are therefore

$$(u, w_j), \quad (-u, w_j).$$

Hence the adjacent sum between Q_{w_j} and $Q_{w_{j+1}}$ is

$$(-u, w_j) + (u, w_{j+1}) = (0, w_j + w_{j+1}).$$

This also holds for the closing edge, so the sums between segments belong to $\{0\} \times S_W$. Together with the sums within segments, this gives (4.2). An edge with sum zero in W yields an edge with sum zero in the new cycle. $\square$

4.2 Low-dimensional constructions

The following ordering is a translate of the prime-field construction in She [2, Remark 1.3].

Lemma 4.3. For every odd prime p , $\mathbb{F}_p$ has a cyclic ordering containing an edge with sum zero and having all adjacent sums in $\{0, 1, -1\}$.

Proof. Let $h = 1/2 \in \mathbb{F}_p$. For integers $1 \leq i \leq p$, define

$$x_i = \begin{cases} i - h, & i \text{ is odd,} \\ 1 - i - h, & i \text{ is even.} \end{cases} \quad (4.4)$$

If i and j have the same parity and $x_i = x_j$, then $i - j \equiv 0 \pmod{p}$, so $i = j$. If i is odd, j is even, and $x_i = x_j$, then $i + j \equiv 1 \pmod{p}$. The odd integer $i + j$ lies in $[3, 2p-1]$, but the only integer in this interval congruent to one modulo p is the even integer $p+1$. This is a contradiction. Thus (x_i) runs through the entire field.

If $i < p$ is odd, then $i+1$ is even, and

$$x_i + x_{i+1} = (i - h) + (1 - (i+1) - h) = -2h = -1.$$

If $i < p$ is even, then $i + 1$ is odd, and

$$x_i + x_{i+1} = (1 - i - h) + (i + 1 - h) = 2 - 2h = 1.$$

Since p is odd, $x_p = p - h = -h$, whereas $x_1 = 1 - h = h$. The closing edge has sum

$$x_p + x_1 = (-h) + (1 - h) = 1 - 2h = 0.$$

Hence all adjacent sums belong to $\{0, 1, -1\}$, and the closing edge has sum zero. $\square$

We next construct two-dimensional orderings using only the labels $0, e_1, e_2$.

Lemma 4.4. For every odd prime $p \geq 5$, $\mathbb{F}_p^2$ has a cyclic ordering whose adjacent sums belong to

$$\{0, e_1, e_2\}, \quad (4.5)$$

and which contains an edge with sum zero.

Proof. For $p = 5$, represent the vector (x, y) by the integer $x + 5y$, where $0 \leq x, y < 5$. Vector addition is performed coordinatewise modulo five. Take the cyclic ordering

$$(0, 5, 21, 9, 22, 8, 23, 12, 18, 13, 17, 14, 16, 19, 11, 15, 10, 20, 6, 24, 7, 3, 2, 4, 1). \quad (4.6)$$

This list contains all 25 vertices. The integers representing its cyclic adjacent sums are

$$(5, 1, 0, 1, 0, 1, 5, 0, 1, 0, 1, 0, 5, 0, 1, 0, 5, 1, 0, 1, 5, 0, 1, 0, 1). \quad (4.7)$$

The integers $0, 1, 5$ represent $0, e_1, e_2$, respectively. Thus every adjacent sum is allowed, and the ordering contains an edge with sum zero.

For $p \geq 7$, write $p = 2m + 1$, where $m \geq 3$. We index the vertices of each path starting at zero and compute coordinates in $\mathbb{F}_p$.

On the line with second coordinate zero, define the path

$$P_0 = ((0, 0), (1, 0), (-1, 0), \dots, (m, 0), (-m, 0)). \quad (4.8)$$

More precisely, $P_0[0] = (0, 0)$, $P_0[2r - 1] = (r, 0)$, and $P_0[2r] = (-r, 0)$ for $1 \leq r \leq m$. For $1 \leq k \leq m$, define the sequence of $2p$ vertices

$$Q_k[2r] = (m - r, k), \quad Q_k[2r + 1] = (m + 2 + r, -k), \quad 0 \leq r < p. \quad (4.9)$$

Concatenate these sequences to form

$$P = P_0 Q_1 Q_2 \cdots Q_m. \quad (4.10)$$

The path P_0 runs through the line $y = 0$, while the even and odd positions of Q_k run through the lines $y = k$ and $y = -k$, respectively. These lines are pairwise disjoint and cover the entire space, so

$$|P| = p + 2pm = p(2m + 1) = p^2.$$

Thus P is a spanning path. The adjacent sums within P_0 alternate between e_1 and 0 . Within Q_k , we have

$$Q_k[2r] + Q_k[2r + 1] = (2m + 2, 0) = e_1, \quad 0 \leq r < p,$$

and

$$Q_k[2r + 1] + Q_k[2r + 2] = (2m + 1, 0) = 0, \quad 0 \leq r < p - 1.$$

The last vertex of P_0 is $(-m, 0)$ and the first vertex of Q_1 is $(m, 1)$; their sum is e_2 . Since

$$Q_k[2p - 1] = (m + 1, -k), \quad Q_{k+1}[0] = (m, k + 1),$$

each edge between consecutive sequences for $k < m$ has sum $(2m+1, 1) = e_2$. Hence every edge of the path has an allowed sum. Its initial and terminal vertices are

$$S = (0, 0), \quad T = (m+1, m+1). \quad (4.11)$$

The edge switches. We modify P by removing the edges listed in Table 3. Write each such edge as $C = \{u_C, v_C\}$; the endpoint names specify how the new edges will be attached.

Table 3: Edges to be removed and their endpoints

Removed edge C	u_C	v_C	Positions in the sequences
A	(m, m)	$(m+2, m+1)$	$Q_m[0], Q_m[1]$
J	$(m-1, m+1)$	$(m+3, m)$	$Q_m[2p-5], Q_m[2p-6]$
$B_k, 1 \leq k \leq m-1$	$(m-2, -k)$	$(2-m, k)$	$Q_k[2p-7], Q_k[2p-6]$
Z	$(m-2, 0)$	$(2-m, 0)$	$P_0[p-6], P_0[p-5]$
F	$(m-2, 1)$	$(2-m, -1)$	$Q_1[4], Q_1[3]$
$L_k, 2 \leq k \leq m$	$(m-k, k)$	$(1-m+k, -k)$	$Q_k[2k], Q_k[2k+1]$
H	$(-1, m+1)$	$(1, m)$	$Q_m[p-4], Q_m[p-3]$
D	$(-1, m+2)$	$(1, m-1)$	$Q_{m-1}[p-4], Q_{m-1}[p-3]$
$E_k, 1 \leq k \leq m-1$	$(0, -k)$	$(0, k)$	$Q_k[p-2], Q_k[p-1]$

Each pair of positions in Table 3 is adjacent and lies within its array. The endpoint orders of J and F along the original path are (v_J, u_J) and (v_F, u_F) , respectively; all other removed edges have order (u_C, v_C) .

For the new edges, order the removed edges as follows:

$$(C_1, \dots, C_K) = (A, J, B_{m-1}, B_{m-2}, \dots, B_1, Z, F, L_2, L_3, \dots, L_m, H, D, E_{m-1}, E_{m-2}, \dots, E_1), \quad (4.12)$$

where $K = 3m+3$. Delete these undirected edges from P and add

$$\{T, u_{C_1}\}, \quad \{v_{C_i}, u_{C_{i+1}}\} \quad (1 \leq i < K), \quad \{v_{C_K}, S\}. \quad (4.13)$$

The sum labels of the new edges are listed in Table 4.

Table 4: Adjacent sums on the new edges

New edge endpoints	Coordinate sum	Sum label
T, u_A	$(2m+1, 2m+1)$	0
v_A, u_J	$(2m+1, 2m+2)$	e_2
$v_J, u_{B_{m-1}}$	$(2m+1, 1)$	e_2
$v_{B_k}, u_{B_{k-1}} \quad (2 \leq k \leq m-1)$	$(0, 1)$	e_2
v_{B_1}, u_Z	$(0, 1)$	e_2
v_Z, u_F	$(0, 1)$	e_2
v_F, u_{L_2}	$(0, 1)$	e_2
$v_{L_k}, u_{L_{k+1}} \quad (2 \leq k < m)$	$(0, 1)$	e_2
v_{L_m}, u_H	$(0, 1)$	e_2
v_H, u_D	$(0, 2m+2)$	e_2
$v_D, u_{E_{m-1}}$	$(1, 0)$	e_1
$v_{E_k}, u_{E_{k-1}} \quad (2 \leq k \leq m-1)$	$(0, 1)$	e_2
v_{E_1}, S	$(0, 1)$	e_2

By Table 4, every new edge has sum in $\{0, e_1, e_2\}$, and $\{T, u_A\}$ has sum zero.

To check the degrees after these switches, first note that the removed edges occur along P in the order

$$Z, F, E_1, B_1, (L_k, E_k, B_k)_{k=2}^{m-2}, D, L_{m-1}, E_{m-1}, B_{m-1}, A, H, L_m, J. \quad (4.14)$$

The starting positions of the removed edges in the respective sequences are $3, p-2, 2p-7$ in Q_1 ; $2k, p-2, 2p-7$ in Q_k for $2 \leq k \leq m-2$; $p-4, p-3, p-2, 2p-7$ in Q_{m-1} ; and $0, p-4, p-1, 2p-6$ in Q_m . They satisfy

$$\begin{aligned} 3 &< p-2 < 2p-7 && \text{in } Q_1, \\ 2k &< p-2 < 2p-7 && \text{for } 2 \leq k \leq m-2, \\ p-4 &< p-3 < p-2 < 2p-7 && \text{in } Q_{m-1}, \\ 0 &< p-4 < p-1 < 2p-6 && \text{in } Q_m. \end{aligned} \tag{4.15}$$

The starting positions within each array are distinct, and different arrays have disjoint vertex sets. Hence all removed edges are distinct. The edges D, L_{m-1}, E_{m-1} are three consecutive edges, with shared endpoints satisfying

$$v_D = u_{L_{m-1}}, \quad v_{L_{m-1}} = u_{E_{m-1}}.$$

Let $r(z)$ be the number of occurrences of vertex z among the removed edge endpoints. None of the removed edges contains S or T , so after deletion the degree of z is

$$2 - \mathbf{1}_{\{S, T\}}(z) - r(z).$$

The edges in (4.13) contribute once for each occurrence of a removed edge endpoint and once at each of S and T . The number of occurrences among the new edge endpoints is $r(z) + \mathbf{1}_{\{S, T\}}(z)$, so every vertex of the resulting multigraph has degree two.

The resulting segments.

For a removed edge C , let C^- and C^+ denote the endpoints occurring earlier and later along the original path. Only F and J satisfy $C^- = v_C$ and $C^+ = u_C$; every other removed edge satisfies $C^- = u_C$ and $C^+ = v_C$. After these edges are removed, each component is a segment of the original path between consecutive removed edges; the first segment starts at S and the last ends at T . Following the order in (4.14), name the segments as in Table 5.

Table 5: Path segments after the cuts

Path segment	Endpoints along the original path	Endpoint notation
O	S to Z^-	S to u_Z
R	Z^+ to F^-	v_Z to v_F
X_1	F^+ to E_1^-	u_F to u_{E_1}
Y_1	E_1^+ to B_1^-	v_{E_1} to u_{B_1}
$I_k, 2 \leq k \leq m-2$	B_{k-1}^+ to L_k^-	$v_{B_{k-1}}$ to u_{L_k}
$X_k, 2 \leq k \leq m-2$	L_k^+ to E_k^-	v_{L_k} to u_{E_k}
$Y_k, 2 \leq k \leq m-2$	E_k^+ to B_k^-	v_{E_k} to u_{B_k}
U	B_{m-2}^+ to D^-	$v_{B_{m-2}}$ to u_D
V	D^+ to L_{m-1}^-	v_D to $u_{L_{m-1}}$
X_{m-1}	L_{m-1}^+ to E_{m-1}^-	$v_{L_{m-1}}$ to $u_{E_{m-1}}$
Y_{m-1}	E_{m-1}^+ to B_{m-1}^-	$v_{E_{m-1}}$ to $u_{B_{m-1}}$
T_0	B_{m-1}^+ to A^-	$v_{B_{m-1}}$ to u_A
T_A	A^+ to H^-	v_A to u_H
T_H	H^+ to L_m^-	v_H to u_{L_m}
T_J	L_m^+ to J^-	v_{L_m} to v_J
T_T	J^+ to T	u_J to T

Here V and X_{m-1} are singleton segments, since

$$v_D = u_{L_{m-1}} = (1, m-1), \quad v_{L_{m-1}} = u_{E_{m-1}} = (0, 1-m).$$

There are $3m + 4 = K + 1$ segments. If the starting positions of the removed edges along the original path are $a_1 < \dots < a_K$, then the position intervals of the segments are

$$[0, a_1], [a_1 + 1, a_2], \dots, [a_{K-1} + 1, a_K], [a_K + 1, p^2 - 1].$$

These intervals are pairwise disjoint and cover all vertices. Consecutive removed edges give singleton intervals. Every edge within a segment is retained from the original path.

Reconnection and connectivity.

Write $B \longleftrightarrow B'$ when a new edge joins the designated endpoints of two path segments. Six of the new edges form the following two paths:

$$\begin{aligned} I_{m-1}^* &= U \longleftrightarrow T_H \longleftrightarrow X_{m-1} \longleftrightarrow V, \\ I_m^* &= T_0 \longleftrightarrow T_T \longleftrightarrow T_A \longleftrightarrow T_J. \end{aligned} \quad (4.16)$$

Their connecting edges and endpoints are listed in Table 6.

Table 6: Connecting edges and endpoints of the two paths

Path	Connecting edges	Endpoints
I_{m-1}^*	$\{u_D, v_H\}, \{u_{L_m}, v_{L_{m-1}}\}, \{u_{E_{m-1}}, v_D\}$	$v_{B_{m-2}}, u_{L_{m-1}}$
I_m^*	$\{u_A, T\}, \{u_J, v_A\}, \{u_H, v_{L_m}\}$	$v_{B_{m-1}}, v_J$

Each intermediate segment is joined at its two endpoint positions. The original segments used in these two paths are pairwise disjoint.

Set $I_{m-1} = I_{m-1}^*$ and $I_m = I_m^*$, retaining the original definition of I_k for $2 \leq k \leq m-2$. For every $2 \leq k \leq m-1$, the path I_k has endpoints $v_{B_{k-1}}, u_{L_k}$, while I_m has endpoints $v_{B_{m-1}}, v_J$. The remaining new edges join the segments into three types of paths connecting endpoints of the Y_i :

$$\begin{aligned} Y_1 &\longleftrightarrow O \longleftrightarrow I_2 \longleftrightarrow R \longleftrightarrow X_1 \longleftrightarrow Y_2, \\ Y_j &\longleftrightarrow I_{j+2} \longleftrightarrow X_{j+1} \longleftrightarrow Y_{j+2} \quad (1 \leq j \leq m-3), \\ Y_{m-2} &\longleftrightarrow I_m \longleftrightarrow Y_{m-1}. \end{aligned} \quad (4.17)$$

The five new edges in the first type of connecting path are, in order,

$$\{v_{E_1}, S\}, \quad \{v_{B_1}, u_Z\}, \quad \{v_F, u_{L_2}\}, \quad \{v_Z, u_F\}, \quad \{v_{E_2}, u_{E_1}\}.$$

These edges join O, I_2, R, X_1 successively, using both endpoints of each path, and connect the v_E endpoints of Y_1, Y_2 . When $m = 3$, $I_2 = I_{m-1}^*$, and its endpoints are given by the same formula.

For a connecting path of the second type, the three new edges are

$$\{v_{B_{j+1}}, u_{B_j}\}, \quad \{v_{L_{j+1}}, u_{L_{j+2}}\}, \quad \{v_{E_{j+2}}, u_{E_{j+1}}\}.$$

They use both endpoints of I_{j+2}, X_{j+1} and join the u_B endpoint of Y_j to the v_E endpoint of Y_{j+2} .

The two new edges in the third type of connecting path are

$$\{v_{B_{m-1}}, u_{B_{m-2}}\}, \quad \{v_J, u_{B_{m-1}}\}.$$

These edges join the u_B endpoints of Y_{m-2}, Y_{m-1} through I_m , using both of its endpoints. Together, the connecting paths and the paths in (4.16) use

$$5 + 3(m-3) + 2 + 6 = 3m + 4 = K + 1$$

new edges, exactly those in (4.13). Every original path segment occurs in one of the paths in (4.16)–(4.17), and the connecting paths have pairwise disjoint interiors.

To prove connectivity, contract each Y_i to vertex i and each connecting path in (4.17) to an edge. Let $M = m - 1 \geq 2$. The resulting graph has vertices $1, \dots, M$ and edges

$$\{1, 2\}, \quad \{j, j+2\} \quad (1 \leq j \leq M-2), \quad \{M-1, M\}. \quad (4.18)$$

The edges $\{j, j+2\}$ form a path on the odd vertices and a path on the even vertices. The edges $\{1, 2\}$ and $\{M-1, M\}$ join the endpoints with the smallest and largest indices, respectively, so the graph is connected. When $M = 2$, these connections give two parallel edges, corresponding to two connecting paths with disjoint interiors. Contraction preserves the number of connected components, so the graph before contraction is also connected.

The graph has $p^2 > 2$ vertices, each of degree two. It is therefore a Hamiltonian cycle.

Choose a starting vertex and a direction along this cycle to obtain the required cyclic ordering. Both the retained and the new edges have sums in $\{0, e_1, e_2\}$, and $\{T, u_A\}$ has sum zero. $\square$

In characteristic three, we use explicit orderings in dimensions one, two, and three.

Lemma 4.5. For $r = 1, 2, 3$, there are cyclic orderings of $\mathbb{F}_3^r$ satisfying Theorem 4.1. The adjacent sums of the three-dimensional ordering belong to $\{0, e_1, e_2, e_3\}$.

Proof. Represent a vector by the integer

$$x_1 + 3x_2 + \dots + 3^{r-1}x_r, \quad x_i \in \{0, 1, 2\},$$

corresponding to its coordinates $(x_1, \dots, x_r)$. Addition of vectors is performed coordinatewise modulo three.

In dimension one, take $(0, 1, 2)$. Its cyclic adjacent sums are $(1, 0, 2)$ in order, where the integer two represents $-e_1$.

In dimension two, take

$$(0, 1, 2, 4, 7, 8, 5, 6, 3). \quad (4.19)$$

The integers representing its cyclic adjacent sums are

$$(1, 0, 3, 2, 3, 1, 2, 0, 3). \quad (4.20)$$

The ordering contains all nine vertices, every adjacent sum belongs to $\{0, e_1, -e_1, e_2\}$, and an edge has sum zero.

In dimension three, take

$$(0, 1, 2, 10, 18, 9, 19, 20, 11, 22, 15, 21, 16, 5, \\ 8, 7, 14, 26, 13, 24, 12, 25, 23, 17, 4, 6, 3). \quad (4.21)$$

The integers representing its twenty-seven adjacent sums are, in order,

$$(1, 0, 9, 1, 0, 1, 9, 1, 3, 1, 0, 1, 9, 1, \\ 3, 9, 1, 0, 1, 0, 1, 9, 1, 9, 1, 0, 3). \quad (4.22)$$

The ordering in (4.21) contains all 27 vertices. The integers 0, 1, 3, 9 in (4.22) represent $0, e_1, e_2, e_3$, respectively, and the ordering contains an edge with sum zero. $\square$

4.3 Extension to arbitrary dimensions

Proof of Theorem 4.1. First suppose $p \geq 5$. If $n = 2t + 1$ is odd, write the space as the direct sum

$$\mathbb{F}_p^n = \langle e_1 \rangle \oplus \bigoplus_{j=1}^t \langle e_{2j}, e_{2j+1} \rangle.$$

Use Lemma 4.3 on the first factor and Lemma 4.4 on each two-dimensional factor. Repeated application of Lemma 4.2 gives a cyclic ordering whose labels belong to Λ_n and which contains an edge with sum zero.

If $n = 2t$ is even, decompose the space into t two-dimensional coordinate factors. Apply Lemma 4.4 to each factor and then Lemma 4.2. The resulting cyclic ordering has labels in $\{0, e_1, \dots, e_n\}$ and contains an edge with sum zero.

Now suppose $p = 3$. Set

$$r = 1 + ((n - 1) \bmod 3) \in \{1, 2, 3\}, \quad t = (n - r)/3 \geq 0.$$

Use the corresponding ordering from Lemma 4.5 on the first r coordinates, followed by t three-dimensional orderings whose labels are zero or standard basis vectors. Applying Lemma 4.2 gives labels in Λ_n and preserves an edge with sum zero. This proves the theorem. $\square$

Corollary 4.6 (Cyclic orderings with fewer labels). If $p \geq 5$ and n is a positive even integer, or if $p = 3$ and n is a positive multiple of three, then $\mathbb{F}_p^n$ has a cyclic ordering containing an edge with sum zero, all of whose adjacent sums belong to

$$\Lambda_n^+ = \{0, e_1, \dots, e_n\}.$$

The superscript $+$ records the omission of $-e_1$ from Λ_n .

Proof. Under these conditions, the construction in Theorem 4.1 uses only two- or three-dimensional cyclic orderings whose labels are zero or standard basis vectors. The result follows from Lemma 4.2. $\square$

5 Primitive adjacent sums in finite fields

We apply the cyclic orderings of Section 4 to finite fields. We first treat odd characteristic, beginning with the field of order nine.

Lemma 5.1. Theorem 1.1 holds when $q = 9$, with four distinct primitive values among the adjacent sums.

Proof. The polynomial $X^2 + 1$ has no root in $\mathbb{F}_3$, so

$$\mathbb{F}_9 = \mathbb{F}_3[\theta], \quad \theta^2 = -1.$$

For $a, b \in \{1, -1\}$, let $z = a + b\theta$. Then

$$z^2 = 2ab\theta, \quad z^4 = -1, \quad z^8 = 1.$$

Each of these four elements z therefore has order eight. The remaining nonzero elements $1, -1, \theta, -\theta$ have orders 1, 2, 4, 4, respectively. Hence

$$\mathcal{P}_9 = \{1 + \theta, 1 - \theta, -1 + \theta, -1 - \theta\}. \quad (5.1)$$

Consider the cyclic ordering

$$(0, 1 - \theta, -\theta, -1 - \theta, -1, \theta - 1, \theta, 1, 1 + \theta). \quad (5.2)$$

Its coordinates in the basis $1, \theta$ are

$$(0, 0), (1, -1), (0, -1), (-1, -1), (-1, 0), (-1, 1), (0, 1), (1, 0), (1, 1),$$

so the sequence lists every element of $\mathbb{F}_9^2$ exactly once. The cyclic adjacent sums are listed in Table 7.

Table 7: Adjacent sums in the cyclic ordering of $\mathbb{F}_9$

Adjacent endpoints	Adjacent sum
$0, 1 - \theta$	$1 - \theta$
$1 - \theta, -\theta$	$1 + \theta$
$-\theta, -1 - \theta$	$-1 + \theta$
$-1 - \theta, -1$	$1 - \theta$
$-1, \theta - 1$	$1 + \theta$
$\theta - 1, \theta$	$-1 - \theta$
$\theta, 1$	$1 + \theta$
$1, 1 + \theta$	$-1 + \theta$
$1 + \theta, 0$	$1 + \theta$

All sums lie in the set in (5.1), and each of the four values occurs. $\square$

An explicit ordering of the field of order nine is also given in [1, Remark 3.5(a)].

Proposition 5.2. For every odd prime power $q = p^n > 7$, there is a cyclic ordering in which every adjacent sum is primitive and at most $n + 2$ distinct sums occur.

Proof. The case $q = 9$ follows from Lemma 5.1. We may therefore assume that $q \neq 9$. Proposition 2.4 gives $c - d, c, c + d \in \mathcal{P}_q$ with $d \neq 0$. By Corollary 3.5,

$$\text{span}_{\mathbb{F}_p}(\mathcal{P}_q - c) = \mathbb{F}_q. \quad (5.3)$$

Set $b_1 = d$ and extend it to a basis $b_1, \dots, b_n$ by choosing the remaining vectors from $\mathcal{P}_q - c$. Then

$$c, c - b_1, c + b_1, c + b_2, \dots, c + b_n \in \mathcal{P}_q. \quad (5.4)$$

Let $(x_0, \dots, x_{q-1})$ be the cyclic ordering in Theorem 4.1. Let $B : \mathbb{F}_p^n \rightarrow \mathbb{F}_q$ be the linear isomorphism satisfying $B(e_i) = b_i$. Then

$$F(x) = \frac{c}{2} + B(x) \quad (5.5)$$

is an affine bijection. Define

$$a_j = F(x_j) = \frac{c}{2} + \sum_{i=1}^n x_{j,i} b_i. \quad (5.6)$$

The resulting sequence lists every element of $\mathbb{F}_q$ exactly once. For cyclic indices $0 \leq j < q$,

$$a_j + a_{j+1} = c + B(x_j + x_{j+1}). \quad (5.7)$$

The adjacent sums $0, e_1, -e_1, e_i$ in the original ordering become $c, c + b_1, c - b_1, c + b_i$, respectively. These elements are primitive by (5.4), so every cyclic adjacent sum is primitive and at most $n + 2$ distinct sums occur. $\square$

We turn to characteristic two, where a cyclic Gray code supplies the required ordering.

Theorem 5.3 (Primitive normal basis theorem [7, 8]). For every finite field extension $\mathbb{F}_{p^n}/\mathbb{F}_p$, there exists a primitive element α such that

$$\alpha, \alpha^p, \dots, \alpha^{p^{n-1}} \quad (5.8)$$

form an $\mathbb{F}_p$ -basis. Since $\gcd(p^i, q - 1) = 1$, every α^{p^i} is primitive.

Proposition 5.4. If $q = 2^n > 7$, then $\mathbb{F}_q$ has a cyclic ordering in which every adjacent sum is primitive and at most n distinct sums occur.

Proof. Theorem 5.3 gives a primitive element α such that the elements

$$b_i = \alpha^{2^i}, \quad 0 \leq i < n, \quad (5.9)$$

form an $\mathbb{F}_2$ -basis. Since $\gcd(2^i, 2^n - 1) = 1$, every b_i is primitive.

For $0 \leq j < 2^n$, write $j = \sum_{i=0}^{n-1} \varepsilon_i(j) 2^i$ and put $\varepsilon_n(j) = 0$. Define the binary reflected Gray code and the corresponding field elements by

$$g_i(j) = \varepsilon_i(j) + \varepsilon_{i+1}(j) \pmod{2}, \quad a_j = \sum_{i=0}^{n-1} g_i(j) b_i. \quad (5.10)$$

The inverse coordinate transformation is

$$\varepsilon_i(j) = \sum_{k=i}^{n-1} g_k(j) \pmod{2}. \quad (5.11)$$

Thus $g(j)$ lists every vector of $\mathbb{F}_2^n$ once, and (a_j) is an ordering of $\mathbb{F}_q$.

For $0 \leq j < 2^n - 1$, let r be the number of trailing ones in the binary expansion of j . Consecutive Gray codewords differ only in bit r , so

$$g_i(j) + g_i(j+1) = \begin{cases} 1, & i = r, \\ 0, & i \neq r, \end{cases} \quad a_j + a_{j+1} = b_r \in \mathcal{P}_q. \quad (5.12)$$

Since $g(0) = 0$ and $g(2^n - 1) = (0, \dots, 0, 1)$, the closing edge satisfies

$$a_{2^n-1} + a_0 = b_{n-1} \in \mathcal{P}_q. \quad (5.13)$$

All adjacent sums belong to $\{b_0, \dots, b_{n-1}\}$, proving the proposition. $\square$

Together with Proposition 5.2, this proves the existence assertion and the basic bounds in Theorem 1.1.

6 Allowed sum sets and lower bounds

Let $V = \mathbb{F}_p^n$ and $S \subseteq V$. We first consider odd characteristic.

Theorem 6.1. Let p be an odd prime and $n \geq 1$. If

$$\text{Aff}(S) = V \quad (6.1)$$

and there exist $c, d \in V$ such that

$$d \neq 0, \quad c - d, c, c + d \in S, \quad (6.2)$$

then V has a cyclic ordering in which every adjacent sum belongs to S and at most $n+2$ distinct sums occur.

Proof. By (6.1)–(6.2), we can set $b_1 = d$ and extend it to a basis $b_1, \dots, b_n$ by choosing the remaining vectors from $S - c$. Use the ordering in Theorem 4.1, define $B(e_i) = b_i$, and apply the affine map $x \mapsto c/2 + B(x)$. By (5.7), all adjacent sums belong to

$$S_0 = \{c, c - b_1, c + b_1, c + b_2, \dots, c + b_n\} \subseteq S, \quad (6.3)$$

which proves the assertion. $\square$

Theorem 6.2. Let p be an odd prime, and suppose that

$$(p \geq 5 \text{ and } 2 \mid n) \quad \text{or} \quad (p = 3 \text{ and } 3 \mid n). \quad (6.4)$$

If $S \subseteq \mathbb{F}_p^n$ affinely spans the whole space, then there is a cyclic ordering in which every adjacent sum belongs to S and at most $n + 1$ distinct sums occur.

Proof. By Corollary 4.6, there is a cyclic ordering whose adjacent sums belong to

$$\Lambda_n^+ = \{0, e_1, \dots, e_n\} \quad (6.5)$$

Choose any $c \in S$, choose a basis $b_1, \dots, b_n$ from $S - c$, and define $B(e_i) = b_i$. The affine map $x \mapsto c/2 + B(x)$ sends the adjacent sums into

$$\{c, c + b_1, \dots, c + b_n\} \subseteq S, \quad (6.6)$$

which proves the bound. $\square$

The affine-spanning condition and lower bounds below are special cases of Lev's results on sums along Hamiltonian cycles [9].

Theorem 6.3. Let p be an odd prime, and let $(a_0, \dots, a_{q-1})$ be a cyclic ordering of $V = \mathbb{F}_p^n$, where $q = p^n$. Put $s_j = a_j + a_{j+1}$ and $T = \{s_j : 0 \leq j < q\}$. Then $\text{Aff}(T) = V$, and consequently

$$|T| \geq \max\{n + 1, 3\}. \quad (6.7)$$

Proof. Let $U = \text{span}_{\mathbb{F}_p} \{s - s' : s, s' \in T\}$. For cyclic indices,

$$s_{j+1} - s_j = a_{j+2} - a_j \in U. \quad (6.8)$$

Since $\gcd(2, q) = 1$, the sequence of indices obtained by successively adding two visits every index. Hence every $a_j - a_0$ belongs to U . As $\{a_j - a_0\} = V$, it follows that $U = V$. Since $U = \text{span}(T - s_0)$, we have $\text{Aff}(T) = V$ and $|T| \geq n + 1$.

Suppose that $T \subseteq \{s, t\}$. At the vertex $x = s/2$, the label s gives a loop, and the label t gives at most one distinct neighbor. The predecessor and successor of x in a cyclic ordering must be distinct, a contradiction. Thus $|T| \geq 3$. $\square$

Any cyclic ordering with adjacent sums in S satisfies $\text{Aff}(S) = V$. Together with Theorem 6.2, this gives, under condition (6.4),

$$\begin{aligned} V \text{ has a cyclic ordering with adjacent sums in } S \\ \iff \text{Aff}(S) = V. \end{aligned} \quad (6.9)$$

The minimum number of labels is then $n + 1$. Taking $S = \mathcal{P}_q$ and applying Proposition 3.4 gives the optimal $n + 1$ -label bound in Theorem 1.1. For $n = 1$ and $q > 7$, Proposition 5.2 and (6.7) show that three primitive labels are necessary and sufficient.

In characteristic two, the corresponding spanning condition is linear. Lev [9] determined that the minimum number of distinct adjacent sums in $\mathbb{F}_2^n$, $n \geq 2$, is n .

Theorem 6.4. Let $V = \mathbb{F}_2^n$, $n \geq 2$, and $S \subseteq V$. There is a cyclic ordering in which every adjacent sum belongs to S if and only if

$$\text{span}_{\mathbb{F}_2}(S) = V. \quad (6.10)$$

When this condition holds, the minimum number of labels is n .

Proof. If S spans V , choose a basis $b_0, \dots, b_{n-1}$ from this set and apply the cyclic Gray code construction in the proof of Proposition 5.4. Every adjacent sum is a member of this basis.

Conversely, suppose that the adjacent sum set of a cyclic ordering is $T \subseteq S$, and let $U = \text{span}(T)$. For $1 \leq j < q$,

$$\sum_{i=0}^{j-1} (a_i + a_{i+1}) = a_0 + a_j. \quad (6.11)$$

The right-hand side takes every value in $V \setminus \{0\}$, so $U = V$. Hence S spans V and $|T| \geq n$. $\square$

Taking $S = \mathcal{P}_{2^n}$, the primitive normal basis theorem gives (6.10), so the label bound in Proposition 5.4 is optimal.

7 Construction and complexity

The input consists of a prime p , a positive integer n , and a monic irreducible polynomial $f \in \mathbb{F}_p[X]$ of degree n . We use the complete prime factorization of $N = p^n - 1$,

$$N = \prod_{\nu=1}^w \ell_\nu^{e_\nu}. \quad (7.1)$$

We work in the field

$$\mathbb{F}_q = \mathbb{F}_p[X]/(f), \quad q = p^n > 7, \quad (7.2)$$

and enumerate its elements in the order determined by the base- p encodings of their coordinates in the polynomial basis. The output is a cyclic ordering $A = (a_0, \dots, a_{q-1})$ of $\mathbb{F}_q$.

If the factorization of $q - 1$ is not supplied, obtain it by trial division, whose bit cost is polynomial in q .

In characteristic two, scan the nonzero field elements α in the fixed order, testing primitivity by (7.3) and rank n of the polynomial-basis coordinate vectors of $\alpha, \alpha^2, \dots, \alpha^{2^{n-1}}$ over $\mathbb{F}_2$. Theorem 5.3 guarantees success. Each of at most $q - 1$ candidate tests takes time polynomial in $\log q$. Generate the binary reflected Gray code and its corresponding field elements by (5.10). By the Gray code argument in the proof of Proposition 5.4, the distinct adjacent sums are exactly the n primitive basis vectors.

When $q = 9$, store the nine coordinates in (5.2). Choose a solution of $\theta^2 = -1$ in the given field representation and replace each coordinate (a, b) by $a + b\theta$. If g is a primitive element of order eight, one may take $\theta = g^2$. Then θ has order four, so $1, \theta$ is an $\mathbb{F}_3$ -basis.

For odd characteristic with $q > 7$ and $q \neq 9$, use the construction below.

For a nonzero element z , use the criterion

$$z \in \mathcal{P}_q \iff z^{N/\ell_\nu} \neq 1 \quad (1 \leq \nu \leq w) \quad (7.3)$$

to find the first primitive element g in the fixed order. Construct the power table $G[t] = g^t$ and the discrete logarithm table $L[g^t] = t$ for $0 \leq t < N$. Primitive elements correspond to exponents satisfying $\gcd(t, N) = 1$, squares to even exponents, and, when $3 \mid N$, cubes to exponents divisible by three.

For the dimensions in (6.4), set $c = g$, choose a basis from $\mathcal{P}_q - c$, and use the cyclic ordering in Corollary 4.6. For the remaining dimensions, enumerate r in the fixed order, set $s = 2 - r$, and require

$$r, s \neq 0, \quad r \neq s, \quad L[r] \equiv L[s] \equiv 0 \pmod{2}; \quad (7.4)$$

if $3 \mid N$, also require

$$L[r] \equiv 0 \pmod{3} \quad \text{or} \quad L[s] \equiv 0 \pmod{3} \quad \text{or} \quad L[r] - L[s] \equiv 0 \pmod{3}. \quad (7.5)$$

Such a pair exists by Section 2. Put $k = L[r]$ and $\lambda = L[s]$. For every prime divisor $\ell \mid N$, select the smallest nonnegative residue t_ℓ satisfying

$$t_\ell \not\equiv 0, -k, -\lambda \pmod{\ell}. \quad (7.6)$$

Solve $t \equiv t_\ell \pmod{\ell}$ by the Chinese remainder theorem, choose the smallest nonnegative solution modulo $\text{rad}(N)$, and put

$$c = G[t], \quad d = \frac{c(s-r)}{2}. \quad (7.7)$$

Lemma 2.1 gives $c - d, c, c + d \in \mathcal{P}_q$. Starting with $b_1 = d$, scan $\mathcal{P}_q - c$ in the fixed order and retain each vector outside the span of those already selected, until a basis $b_1, \dots, b_n$ is obtained.

Construct the cyclic ordering from the low-dimensional orderings and product construction in Section 4. When $p \geq 5$, use orderings on two-dimensional coordinate subspaces, preceded by a one-dimensional ordering if n is odd. When $p = 3$, choose $r_0 \in \{1, 2, 3\}$ with $r_0 \equiv n \pmod{3}$, start with the corresponding ordering on the first r_0 coordinates, and combine it with orderings on three-dimensional coordinate subspaces. Start each low-dimensional ordering at the vertex with the smallest encoding and choose its orientation by taking the neighbor with the smaller encoding first. At each product step, use the first zero-sum edge in the current sequence. Denote the resulting cyclic ordering by $X = (x_0, \dots, x_{q-1})$, and apply the affine map (5.6).

Algorithm 1 Cyclic ordering with primitive adjacent sums in odd characteristic

Input: An odd prime p and a positive integer n with $q = p^n > 7$ and $q \neq 9$; a monic irreducible polynomial $f \in \mathbb{F}_p[X]$ of degree n and the complete prime factorization of $N = q - 1$ as in (7.1)–(7.2).

Output: A cyclic ordering $A = (a_0, \dots, a_{q-1})$ of $\mathbb{F}_q$ with $T(A) \subseteq \mathcal{P}_q$ and $|T(A)| \leq n + 2$; under (6.4), $|T(A)| = n + 1$ (Theorem 1.1).

- 1: Select the first primitive element g in the fixed order, using (7.3).
 - 2: Construct $G[t] = g^t$ and $L[g^t] = t$ for $0 \leq t < N$.
 - 3: **if** ($p \geq 5$ and $2 \mid n$) or ($p = 3$ and $3 \mid n$) **then**
 - 4: $c \leftarrow g$.
 - 5: Scan $\mathcal{P}_q - c$ in the fixed order, retaining vectors outside the span of those already selected until a basis $b_1, \dots, b_n$ is obtained.
 - 6: Construct $X = (x_0, \dots, x_{q-1})$ as in Corollary 4.6, using the fixed choices above.
 - 7: **else**
 - 8: Choose the first r in the fixed order for which $s = 2 - r$ satisfies (7.4) and, when $3 \mid N$, (7.5).
 - 9: $(k, \lambda) \leftarrow (L[r], L[s])$.
 - 10: For each prime $\ell \mid N$, choose the smallest nonnegative residue t_ℓ satisfying (7.6).
 - 11: Solve $t \equiv t_\ell \pmod{\ell}$ by the Chinese remainder theorem, taking $0 \leq t < \text{rad}(N)$.
 - 12: $c \leftarrow G[t]$, $d \leftarrow c(s-r)/2$.
 - 13: Starting with $b_1 = d$, extend to a basis by the same scan of $\mathcal{P}_q - c$.
 - 14: Construct $X = (x_0, \dots, x_{q-1})$ as in Theorem 4.1, using the fixed choices above.
 - 15: **end if**
 - 16: Define $B : \mathbb{F}_p^n \rightarrow \mathbb{F}_q$ by $B(e_i) = b_i$ for $1 \leq i \leq n$.
 - 17: Set $a_j \leftarrow c/2 + B(x_j)$ for $0 \leq j < q$, as in (5.6).
 - 18: **return** $A = (a_0, \dots, a_{q-1})$.
-

The search for a primitive element terminates because the multiplicative group is cyclic. The parameter search terminates by Section 2, and the basis extension succeeds by Corollary 3.5. The required cyclic orderings are given by Theorem 4.1 and Corollary 4.6. The affine map $x \mapsto c/2 + B(x)$ is a bijection and transforms adjacent sums into $c + B(x_j + x_{j+1})$. The labels of the ordering in Theorem 4.1 map into $\{c, c - b_1, c + b_1, c + b_2, \dots, c + b_n\}$, and those of the ordering in Corollary 4.6 into $\{c, c + b_1, \dots, c + b_n\}$. Hence the output has primitive adjacent sums, with at most $n + 2$ and $n + 1$ labels, respectively.

Assume that the field representation and factorization are given, and put $w = \omega(q - 1)$. Testing at most $q - 1$ candidates by binary exponentiation requires $O(qw \log q)$ finite-field operations. The power table, discrete logarithm table, and parameter scan require $O(q)$ finite-field

operations. For each prime modulus, at most four residues need to be tested. The Chinese remainder calculation uses integers with $O(\log q)$ bits. Incremental basis extension requires $O(qn^2)$ base-field operations, generating the cyclic ordering in $\mathbb{F}_p^n$ requires $O(nq)$ base-field operations, and the affine map requires $O(nq)$ finite-field operations.

The general construction therefore requires

$$O(qw \log q + nq) \text{ finite-field operations,} \quad O(qn^2) \text{ base-field operations,}$$

with integer arithmetic counted separately. The two tables, the cyclic ordering in $\mathbb{F}_p^n$, and the complete output require a total of $O(nq)$ base-field coordinates of storage; the complete ordering occupies $\Theta(q \log q)$ bits. In characteristic two, once a primitive normal basis is given, generating the binary reflected Gray code requires $O(nq)$ bit operations, and applying the linear map requires $O(nq)$ additions in $\mathbb{F}_q$. For the field of order nine, reading the stored coordinates and substituting θ has constant cost.

Since $w \leq \log_2(q-1)$ and $n \geq 1$, the general odd-characteristic construction has total arithmetic complexity

$$O(q(\log q)^2 + qn^2), \tag{7.8}$$

where operations in $\mathbb{F}_q$ and $\mathbb{F}_p$ are counted at unit cost and integer arithmetic is accounted for separately.

Field arithmetic in the polynomial representation and the Chinese remainder calculation take time polynomial in $\log q$. Including initialization, the algorithm therefore runs in deterministic time polynomial in q in every characteristic.

It remains to determine the minimum number of labels in the other odd-characteristic dimensions and to reduce the cost of parameter selection.

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